Business Question and Visualization Report

Date	24 September 2024
Team ID	PNT2022TMIDxxxxxx
Project Name	FORMULA 1 Race Intelligence Dashboard Analysis Report (2010-2026)
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Sample

1. **What are the monthly sales trends?**
   - *Visualization*: Line chart showing monthly sales trends.
   - *Screenshot of visualisation*
2. **Which products have the highest revenue?**
   - *Visualization*: Bar chart comparing revenue by product
   - *Screenshot of visualisation*
3. **How is customer satisfaction varying across regions?**
   - *Visualization*: Map visualization showing customer satisfaction by region.
   - *Screenshot of visualisation*

Note: Min 8 business question and visualisations Required inform of above

Formula 1 Business Questions and Visualization Plan

Use these questions as the minimum analysis plan. Add screenshots only after the corresponding Tableau worksheet has actually been built. The provided workbook contains no driver-nationality field, so nationality talent-pool analysis requires a separately sourced, documented mapping and is not part of the current workbook-only analysis.

#	Business question	Recommended Tableau visual	Fields / calculation
1	Which drivers accumulate the most championship points across the available seasons?	Horizontal bar chart; SUM(points) by driver; filter season.	driver, points; SUM(points)
2	How do constructor points change season by season?	Line chart; season on Columns, SUM(points) on Rows, team on Color; compare Red Bull Racing and Mercedes.	season, team, points; SUM(points)
3	Which tyre compounds are used most often in the lap-level sample?	Bar chart; COUNT(lap_number) by tyre_compound; optionally filter season/driver.	tyre_compound; COUNT(lap_number)
4	How does tyre age relate to lap time?	Scatter plot; tyre_age_laps vs lap_time_s, color by compound; inspect outliers and circuit context.	See dataset fields; validate grain and nulls before interpreting.
5	Which countries and circuits appear most often in race-result records?	Use COUNTD(race_id) by country/circuit after validating race_id grain; label as events only if unique IDs are correct.	See dataset fields; validate grain and nulls before interpreting.
6	Which drivers and teams have the most podium finishes?	Filter finish_position to 1, 2, 3 and count results; split by driver/team.	finish_position in 1–3; count results
7	How many DNFs occur by team, season and reason?	Stacked bars or heatmap; SUM(dnf) by team/season; use dnf_reason only where populated.	dnf, dnf_reason, team, season
8	Do wet races or circuit types show different DNF rates?	Compare AVG(dnf) by circuit_type and is_wet_race; show denominator/record count.	See dataset fields; validate grain and nulls before interpreting.
9	How do grid position and finish position relate to positions gained?	Scatter or bar chart; grid_position vs finish_position; use positions_gained as a cross-check.	See dataset fields; validate grain and nulls before interpreting.
10	How do safety-car periods, pit stops and pit-stop duration vary by race?	Scatter/dual view; safety_car_laps, pit_stops and avg_pit_stop_time_s by race_id/season.	See dataset fields; validate grain and nulls before interpreting.